Cryptography Term Project 2025-26

Secure Centralized EV Charging Payment Gateway using Post-Quantum and Lightweight Cryptography

1. Introduction

Increasing adoption of EVs has led to a massive expansion in the EV charging infrastructure. As drivers utilize these public charging stations, they require an efficient payment system. However, this system consisting of a decentralized physical infrastructure combined with a centralized payment processing system is vulnerable to major security challenges. Critical requirements of an EV charging network could include

a. Ensuring secrecy of driver's location, vehicle data
b. Protection against payment fraud, session hijacking
c. Maintaining accurate billing logs

To ensure data confidentiality, prevent tampering, and maintain trust among EV owners and charging station providers, integration of **Blockchain**, **Lightweight Cryptography (LWC)**, and **Quantum Cryptography** is proposed in this system. This system will use blockchain for maintaining transaction records of energy purchases, LWC for fast-encryption on low-power charging kiosks and Quantum Cryptography using Shor's algorithm to identify potential vulnerabilities in the initial key exchanges securing user credentials.

In the subsequent sections we describe the problem statement in detail to design a Centralized EV Charging Payment Gateway by describing the core functionalities, data flow, encryption methods, and technical requirements. Also described are the role of each participating entity, the process of payment initiation, the security infrastructure and the expected challenges

2. System Overview

The aim of Centralized EV Charging Payment Gateway is to simulate a complete digital transaction system for purchasing electric vehicle charging time in a securely and in a verifiable manner. This system will be designed using three physical devices acting as separate entities:

a. **Charging Kiosk:** This is the central processing machine mimicking the automated dispenser at an EV charging station. Responsible for encrypting the station's

franchise information and communicating with the central grid authority and the user.

b. **User Device (EV Owner):** User platform where the EV Owner initiates a charging session by scanning a station QR code, entering their required energy amount, vehicle Mobile ID (VMID), and PIN.
c. **Grid Authority Laptop:** Central banking terminal allowing the energy grid administrators to access the transaction and billing data.

The system consists of the following critical processes:

- **Grid Registration:** There are 3 central energy providers in our system (e.g., Tata Power, Adani, ChargePoint). Each provider has 3 regional zones identified by a unique zone code. Station franchises and users will register their accounts under these zones.

- **Franchise Registration:** Charging station owners (Franchises) must register with the Grid by providing their franchise name, zone code, password, and initial account balance. The Grid generates a 16-digit Franchise ID (FID). FID is formed using their name, time of account creation, and password hashed using the SHA-3 (Keccak-256) algorithm and converted into a 16-digit hexadecimal number. This is unique to every station and shouldn't be shared.
- **User Registration:** EV Owners also need to have an account linked to the Grid. The registration is similar to a franchise, but the Grid also stores a PIN set by the user for authorizing charges. The 16-digit number is referred to as UID. Users use their UID and mobile number to create their Vehicle Mobile ID (VMID).
- **QR Code Generation:** The station kiosk takes its Franchise ID (FID), encrypts it using an LWC algorithm, and generates a QR code on the kiosk screen which upon scanning reveals the encrypted number.
- **Payment Process:** The EV Owner scans the QR code, providing their VMID, required charging amount (in currency), and PIN to initiate the session. This data is hashed and sent to the Charging Kiosk.
- **Transaction Processing:** The Charging Kiosk decrypts the Franchise ID from the QR code, forwards the authorization request to the Grid Authority, and waits for approval to begin dispensing power.
- **Grid Processing:** The Grid decrypts the user credentials and validates the funds, VMID, and PIN. If successful, the transaction is recorded in the blockchain, and funds are transferred to the Franchise.

Integration of Blockchain ensures that each transaction is immutable and transparent, Lightweight Cryptography guarantees fast and secure encryption.

3. Entities Involved

The system is designed to interact between four key entities: the Grid Authority, Franchise, EV Owner, and Charging Kiosk. Each entity has a unique role and responsibility in ensuring the successful processing of a transaction.

3.1 Grid Authority

The Grid Authority is the central governing body responsible for managing the energy accounts of both users and franchises. It acts as the ultimate decision-maker for processing payments, validating credentials, and ensuring funds availability.

Responsibilities of the Grid Authority:

- **Registration:** Registers both EV Owners and Station Franchises.
- **Generating Virtual Franchise ID (VFID):** The franchise enters its FID into the charging kiosk, which uses an LWC algorithm to generate dynamic VFID.

- **Transaction Verification:** Verifies VMID, PIN, and balance before authorizing the charging session.
- **Blockchain Ledger Maintenance:** Stores all valid charging transactions in a secure blockchain ledger.

3.2 Franchise

The Franchise is the business entity operating the physical charging station.

Responsibilities of the Franchise:

- **Providing Grid Details:** The franchise shares their account details including Account Number, Zone Code, and business information.
- **QR Code Generation:** Receives a QR code from the Kiosk containing the hashed VFID to display on the screen.
- **Transaction Confirmation:** Receives a success/failure message from the Grid to physically unlock the charging cable.

3.3 EV Owner

The EV Owner is the individual initiating the charging session by scanning the QR code and entering the payment details.

Responsibilities of the EV Owner:

- **Providing Payment Details:** The user provides their VMID, PIN, and transaction amount during the payment process.
- **Authorizing Payment:** Confirms the payment after scanning the QR code.
- **Receiving Confirmation:** Receives a success/failure message from the Grid through the Charging Kiosk.

3.4 Charging Kiosk

The Charging Kiosk acts as an intermediary between the EV Owner, Franchise, and Grid. It ensures encrypted communication and relays transaction requests to the Grid.

Responsibilities of the Charging Kiosk:

- **Encrypting Franchise ID:** The Charging Kiosk encrypts the Franchise ID to generate the QR code.
- **Relaying Transactions:** Sends transaction requests to the Grid and waits for approval.
- **Providing Confirmation:** Based on the Grid's response, it notifies both the user and franchise regarding payment success/failure.

4. Role of Lightweight Cryptography (LWC)

LWC is critical because EV charging stations and vehicle communication modules often operate on constrained embedded systems. LWC is used for:

- Generating a Virtual Franchise ID (VFID) from the original FID and a timestamp.
- Ensuring low computational overhead to reduce authorization delays at the plug.
- Encrypting Franchise ID in the QR code.
- **Use the ASCON algorithm for lightweight cryptography, designed for secure, high-speed encryption in resource-constrained IoT environments.**

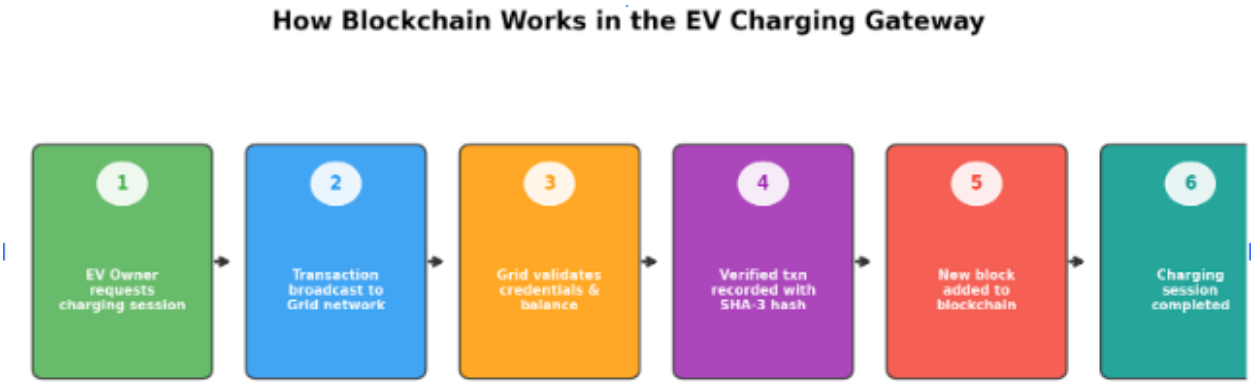
5. Role of Quantum Cryptography

Quantum Cryptography is introduced in the following manner:

- **Simulate Shor's Algorithm to break the RSA-encrypted public key exchange used to transmit the EV Owner's PIN and VMID over the network.**
- This highlights the vulnerability of classical cryptography in modern smart-grid and automotive networks.
- Use Quantum Cryptographic libraries to demonstrate the potential breach.

6. Role of Blockchain

The Grid maintains a Centralized Blockchain Ledger where valid charging sessions are recorded.



Transaction Block Structure:

- Transaction ID (SHA-3 Hash of UID, FID, Timestamp, Amount)
- Previous Block Hash

- Timestamp
- **Dispute/Refund Flag (Boolean flag to handle failed charging sessions: e.g., if payment succeeds but the hardware fails to dispense power, a reverse block must be added).**
- Every valid transaction creates a new block.

This ensures immutability and transparency in EV charging transactions.

Note: Students are expected to handle reasonable edge cases using their own assumptions. Examples include (but are not limited to) account closure mid-session, insufficient balance during transaction processing, or hardware failure after successful payment. Document your assumptions clearly

7. Outcomes of the Project

By the end of this project, the following outcomes need to be achieved:

- A functional centralized EV Charging Payment Gateway mimicking smart-grid transactions.
- Lightweight Cryptography implementation using ASCON.
- Quantum Cryptography Simulation (Shor's Algorithm) to demonstrate public key vulnerabilities.
- Blockchain Integration for secure, immutable energy billing and dispute handling.

8. Further Reading

- ASCON
- SHA-3 Standard
- Shor Algorithm Explained
- Blockchain

9. Deliverables

Submit your source code and a README file as a single .zip or .tar file. Please name your file as **BITSF463_Team_99** (assuming your team number is 99. You can find your team number from the Google sheet). The readme should contain a brief explanation of the project, steps to run the code, and the list of team members. Please submit the solution through the following Google form link.

Google form link for submission of term project solution:

Deadline: 11:59 PM, April 14th, 2026. The exact date and demo schedule will be intimated later. From a team, only one team member needs to do the submission.

For queries regarding the assignment, the students may approach the TAs of the course:

Ashmik Harinkhede (h20250037@hyderabad.bits-pilani.ac.in)

Harsh Patel (h20250179@hyderabad.bits-pilani.ac.in)

Kritii Gupta (f20220757@hyderabad.bits-pilani.ac.in)